CTE 1.1 VIGIL Agent > TRL 3.3

This document is in the context of the core System description and the core Data description. The overall system design slides further describes how these pieces fit together for fine-grained guidance.

VIGIL Agent Performance Analysis

VIGIL depends on multiple interconnected subcomponents, including speech recognition and the visual question answering (VQA) system. The agent receives transcribed audio from the medical generalist, then processes this transcription through the contextual trigger analysis model. This model determines:

1. Whether the generalist is speaking to VIGIL or the patient
2. Whether a state transition is needed
3. Whether the generalist is asking a direct question

Accurate interpretation at this stage is essential because all downstream behavior such as state management, tool use, and guidance generation relies on these determinations.

<u>Time to Response/Time to Guidance (TTR/TTG)</u>

Model	System Prompt	Avg. TTR/TTG (seconds)
Qwen2.5	OriginalPrompt	27.3
Qwen2.5	OriginalToolsPrompt	42.1
Qwen2.5	ReducedToolsPrompt	18.7
Qwen2.5	SimpleToolsPrompt	8.7
Qwen2.5	SimplestToolsPrompt	4.5

* After evaluation, we found that state management can be performed with an additional LLM call each iteration rather than through a separate tool. This change substantially reduces latency. As shown in the last two rows of the table, SimpleToolsPrompt embeds the state tool, while SimplestToolsPrompt omits it; resulting in a 51% decrease in response time.
** We plan on evaluating the same set of system prompts with additional open-source local models including, but not limited to, Llama, Falcon, and Mistral.
*** All prompts are listed at the end of this document.

<u>State Transition Accuracy</u>

State transition accuracy refers to how reliably the agent identifies which internal operational state it should occupy based on user utterances. These transitions correspond to major stages of task execution (for example: idle -> observation -> analysis -> guidance).

During the Dec 1st to 14th user studies, we will be collecting all transcribed text from the participants which will be included in the ROSBAGs. We will also be interviewing all participants regarding their experience with the VIGIL agent.

After the user studies, we will evaluate:

1. Accuracy across a wide variety of vocabulary and speech patterns, collected from many different participants.
2. Performance across natural, unstructured dialogue, not scripted interaction.
3. Situations where multiple transition cues appear in one utterance.
4. Cases where no transition should occur, but the model might falsely trigger.

This evaluation will be run across all system prompts and model variants.

<u>Tool Calling Evaluation</u>

Our tool calling evaluation measures whether the agent invokes the correct tool given the detected user intent and operational state, and whether it provides the correct tool arguments.

During the Dec 1st to 14th user studies, we will be collecting all transcribed text from the participants which will be included in the ROSBAGs. We will also be interviewing all participants regarding their experience with the VIGIL agent.

After the user studies, we will measure:

1. Precision of tool selection across diverse phrasings (e.g., “the gel was applied,” “the probe is on,” “ready for fine grained guidance”).
2. Whether the agent unnecessarily calls tools when conversational responses are sufficient.
3. Robustness to ambiguous or incomplete user statements.
4. Consistency of tool use across different prompts and different base models.

For evaluating whether the agent correctly decides to call the VQA tool, we will use the following metrics and definitions (treating “call VQA tool” as the positive class):

Precision = TP / (TP + FP)
Recall = TP / (TP + FN)
F1 = 2·Precision·Recall / (Precision + Recall)
Overuse (extraneous tool calls) = FP / (TP + FP)
Underuse (missed tool calls) = FN / (TP + FN)

Coverage fraction of queries where the agent chose to call the tool) = (TP + FP) / (TP + FP + TN + FN)

For completeness, we will also track qualitative error types not captured by the above metrics, including:

- Wrong tool invoked (if we are using the State Management as separate tool)
- Incorrect or malformed tool arguments

You can find all 5 system prompts we benchmarked here: https://www.dropbox.com/scl/fo/wrsqsdj9qcp8lrf0bwzc3/ABp8wM73vpux0fpEplbMT7U?rlkey=fyu5onudrvte9kk1gsyzfs8r7&st=u00k1igs&dl=0